Syed Hashim Shah

Quantitative ML & Generative AI Researcher for Finance & Risk Management

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Quantitative researcher from a computational biophysics and engineering background with 3+ years of experience in building ML / generative AI models for financial risk management. Works with non-convex optimization, stochastic processes, simulation, Markov chain Monte Carlo and related methods and loves solving complex problems.

Work Experience

Senior Consultant, Aurora Solutions **Tokyo, Japan**
Quantitative ML & Generative AI Researcher for Finance & Risk Management *Oct 2023 – Present*

- **Generative models from scratch**: VAE, flow-matching and LSTM models for realistic interest-rate and futures scenarios in margin and stress testing. Lead author of **JPX Working Paper Vol. 48** *(2025)*.
- **Synthetic market data**: price histories for instruments with a short trading history, benchmarked against PCA, regression, random forest and XGBoost on distribution and VaR / ES tests.
- **Stress-testing algorithms for GenAI models**: constrained optimization over portfolio sensitivities for realistic scenarios on chosen portfolios; out-of-sample validation on past market crises.
- **Risk-model validation and ML analytics**: VaR and expected-shortfall backtests (Kupiec, ES severity tests), breach analytics, implied-volatility curve classification and anomaly detection.
- **Neural solvers for model calibration**: simulation-based inference for option-pricing models (Heston, SLV), with Monte Carlo and convex optimization for parameter fitting.

Engineer and Researcher, GIKI, Faculty of Materials & Chemical Engineering **Topi, Pakistan**
Research & Teaching *Sep 2016 – Mar 2018*

- **Lab setup and teaching**: new electrochemistry and nanotechnology laboratory; lab courses for 400+ students; research on electrochemical nanomaterials.

Education

Kyoto University, Dept. of Biophysics, Graduate School of Science **Kyoto, Japan**
PhD in Computational Biophysics (ongoing) *Oct 2020 – Present*

- All-atom molecular-dynamics simulations of proteins and DNA on supercomputers including **Fugaku**; free-energy calculations and enhanced-sampling methods.
- *Methods:* stochastic simulation, rare-event path methods, high-dimensional optimization, statistical physics, Python.

Nagoya University, Dept. of Physics, Graduate School of Science **Nagoya, Japan**
M.Sc. in Materials Physics (Theoretical Biophysics Lab), GPA 4.00/4.00 *Oct 2018 – Sep 2020*

- Quantum-chemical enhanced sampling and free-energy calculations. *Thesis:* free-energy cost of the proline dihedral transition by quantum MD with replica-exchange umbrella sampling.
- *Focus:* Markov Chain Monte Carlo, Optimization, Statistical Physics, Quantum Physics.

Ghulam Ishaq Khan Institute (GIKI) **Swabi, Pakistan**
B.Sc. in Materials Engineering (Nanotechnology), GPA 3.88/4.00, Gold Medal *Aug 2012 – Jun 2016*

- *Focus:* Engineering Mathematics, Numerical Methods, Statistics, Programming, Nanomaterials.

Technical Skills

- **Generative AI & ML:** Variational autoencoders (incl. conditional), normalizing flows and flow matching, LSTM sequence models, time-series foundation models, anomaly detection, gradient boosting (XGBoost), random forests.
- **Quantitative Risk:** VaR and expected shortfall, backtesting (Kupiec, ES severity tests), stress testing and scenario generation, PCA term-structure models, volatility models (GARCH, HAR), option-model calibration (Heston, SLV, rough volatility).
- **Statistics & Optimization:** Bayesian Inference, Simulation-Based Inference, Markov Chain Monte Carlo, Stochastic Processes, Time-Series Analysis, Inverse Problems, Graph Theory, convex, non-convex and mixed-integer optimization, simulated annealing and replica exchange, high-dimensional optimization.
- **Programming & Computing:** Python (PyTorch, scikit-learn, XGBoost, pandas, NumPy), C++, Linux, Bash, Git, Jupyter, Quarto; HPC on supercomputers (Fugaku).
- **Languages:** English (fluent), Japanese (**JLPT N3**), Urdu (native).

Awards & Achievements

- **MEXT Scholarship**: Fully funded Japanese Government Monbukagakusho Scholarship for Masters and PhD *(Apr 2018 – Sep 2023)*.
- **Lindau Nobel Laureate Meeting**: Fellowship from the Lindau Foundation and Else Kröner-Fresenius-Stiftung *(Jun 2023)*.
- **Merit Scholarship**: Fully funded award covering the entire undergraduate degree at GIKI *(2012)*.
- **Gold Medal**: Best academic performance in B.Sc. Materials Engineering, GIKI *(Jun 2016)*.

Publications & Presentations

- Shah, S. H. (lead author) (2025). *Hypothetical Market Data Scenario Generation Using Generative AI.* **JPX Working Paper Vol. 48**, Japan Exchange Group (JPX) / JSCC.
- Shah, S. H., Khan, M. I., Sarfraz, R., Nazir, R., & Ali, A. (2018). *Polyaniline Enhanced Supercapacitance of Cobalt Hydroxide Nanowires/Carbon Nanotube Containing Polymer Sponge Layered Composite.* **Key Engineering Materials, 778, 175–180.**
- **Presentation & Poster:** *Elucidation of nucleosome sliding mechanism in all-atom detail via MD simulations*, 60th (Sep 2022) and 59th (Nov 2021) Annual Meetings, Biophysical Society of Japan.

Continuous Professional Learning

- **Introduction to Corporate Finance**: Wharton School, University of Pennsylvania.
- **Python and Statistics for Financial Analysis**: Hong Kong University of Science and Technology.
- **Data Visualization with Python**: IBM.
- **Introduction to Stochastic Processes**: HSE University.